

Proactive Deepfake Defense — Code Review Checklist

Target Domain: DCT-Domain Adversarial Watermarking against StarGAN

Verified Metrics Summary

- PSNR: 30.28 dB
 - SSIM: 0.9061
 - Defense Success Rate (SR_mask): 100.00%
 - Hyperparameters: $\epsilon = 0.018$, $\alpha = 0.003$, $T = 20$ steps, JPEG Q = 75
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1. System Architecture & Lifecycle

1. How is the project modularized across `main.py`, `watermark.py`, `jpeg_utils.py`, and `evaluation.py`?
2. What is the exact execution flow when running `main.py`, from dataset batching to metric outputs?
3. How are CelebA and LFW datasets loaded, transformed, and cached during CPU execution?

2. Frequency-Domain Adversarial Optimization

1. How is the spatial RGB image transformed into the 8x8 block DCT domain inside `watermark.py`?
2. How is the binary mid-frequency mask ($2 \leq u + v \leq 6$) applied to restrict perturbations?
3. What is the step-by-step PGD update rule during the $T=20$ iterations through StarGAN?
4. How is tensor clamping enforced post-update to prevent low/high frequency leakage?
5. How is the universal watermark tensor aggregated or averaged across training batches?

3. Loss Mechanics & Hyperparameters

1. What is the exact composite loss function ($L_{\text{total}} = L_{\text{disrupt}} + \lambda L_{\text{reg}}$) used in PGD?
2. How does the regularization term $\lambda = 0.10$ keep high-frequency noise under control?
3. How is target attribute vector c sampled—is it fixed or randomized per batch?

4. JPEG Differentiability & Compression

1. How does `jpeg_utils.py` (DiffJPEG) simulate lossy JPEG compression ($Q=75$) differentiably?
2. Is the watermark injected strictly into the Y channel or across all YCbCr channels?
3. How are standard JPEG quantization tables handled during forward and backward passes?

5. Target Model Integration (StarGAN)

1. How are StarGAN generator weights loaded, and are generator parameters frozen?
2. How is StarGAN's input normalization range $[-1, 1]$ enforced to prevent saturation?

6. Metrics, Thresholds & Visualization

1. How does `evaluation.py` calculate image quality metrics (PSNR, SSIM, MSE) in $[0, 1]$ space?
2. How is the 15% facial border cropping applied before calculating L_2^{mask} disruption?
3. Why is $\tau = 0.005$ selected as the threshold, and how does it relate to $\text{SR}_{\text{mask}} = 100\%$?
4. How does `visualize.py` apply dynamic scaling ($|I_{\text{protected}} - I_{\text{clean}}| \times k$) for noise visibility?

7. Pipeline Safety & Numerical Stability

1. Where in the pipeline are `torch.clamp(..., 0.0, 1.0)` operations enforced for PSNR stability?
2. How does the pipeline handle precision safety during spatial $\rightarrow$ YCbCr $\rightarrow$ 2D-DCT conversions?